Algebra 1 Name_____Answer key____
Unit 7 Date___________________
Lesson 3 Homework Period__________________

For questions 1-5, factor each quadratic expression using the method of your choice.

1. $b^2 - 2b - 63$

$(b-9)(b+7)$

2. $h^2 + 4h - 77$

$(h+11)\,(h-7)$

3. $x^2 - 5x - 50$

$(x-10)(x+5)$

4. $y^2 - 13y + 36$

$(y-9)(y-4)$

5. $x^2 - x - 132$

$(x-12)(x+11)$

For questions 6-7, consider the trinomial $f^2 - 3f - 10$.

6. Complete the area model.

	f	-5
f	f^2	$-5f$
$+2$	$2f$	-10

7. What are the two binomial factors for $f^2 - 3f - 10$?

$(f-5)$ and $(f+2)$

For questions 8-9, consider the trinomial $m^2 - 22m + 120$.

8. Complete the area model.

	m	-10
m	m^2	$-10m$
-12	$-12m$	$+120$

9. What are the two binomial factors for $m^2 - 22m + 120$?

$(m - 10)$ and $(m - 12)$

10. Factor $x^2 - x - 20$ using factoring by grouping.

$(x^2 + 4x) + (-5x - 20)$
$x(x + 4) + (-5)(x + 4)$
$(x - 5)(x + 4)$